IIT Madras

ONLINE DEGREE

Database Management Systems
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Widely Used DBMSs and Course Summarization

Welcome to Module 60 of Database Management Systems course, in the IIT Madras, online BSc program.

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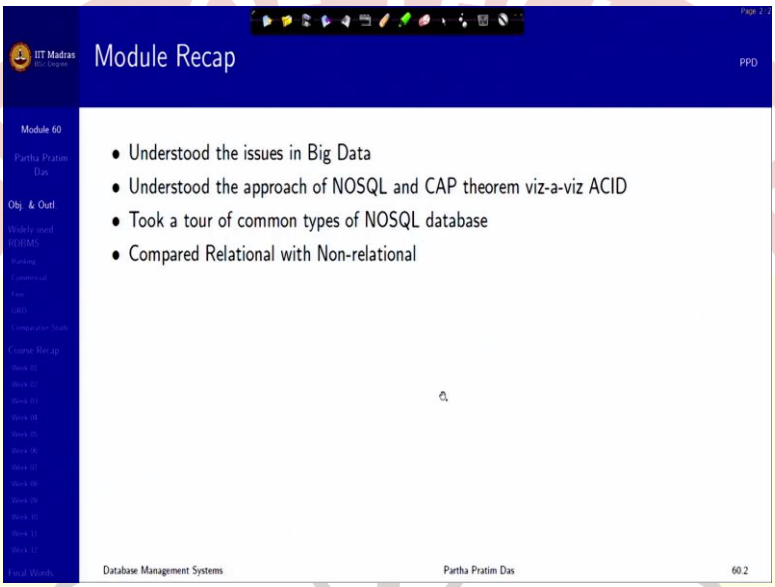

In the last module we talked about big data, which is emerging very strongly these days. And we understood the approach of NOSQL, that is non-relational approach to database modeling, design and implementation. And saw the consequences of the CAP theorem, in contrast to the ACID properties, the BASE properties so to say. We also took a tour of common types of NOSQL databases and compared the relational and non-relational databases.

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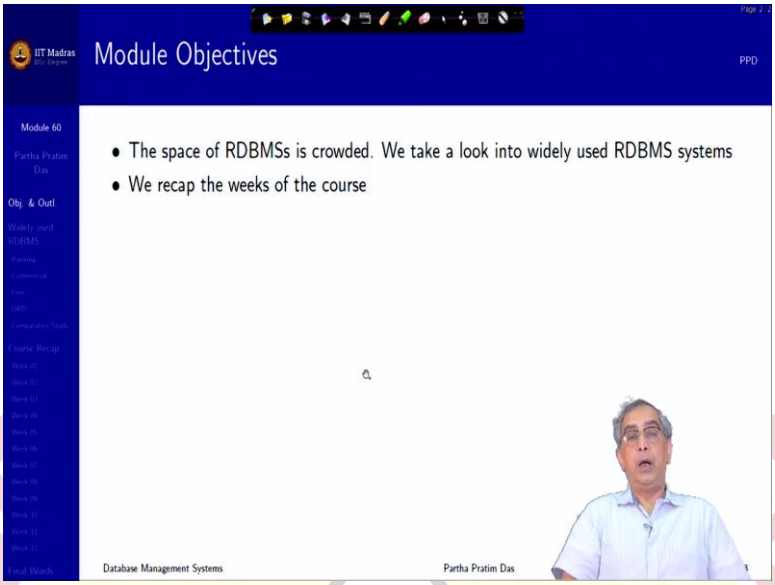

In this module, which is going to be the last module of the course, we would take a look into the widely use RDBMS systems. So, you must understand that this is not for memorizing everything by heart, but to give you an idea of what are the commonly used systems and the probability that when you go to work, he will work with one of them is extremely high. And we of course, recap the weeks of the course very briefly, just to be able to summarize everything.

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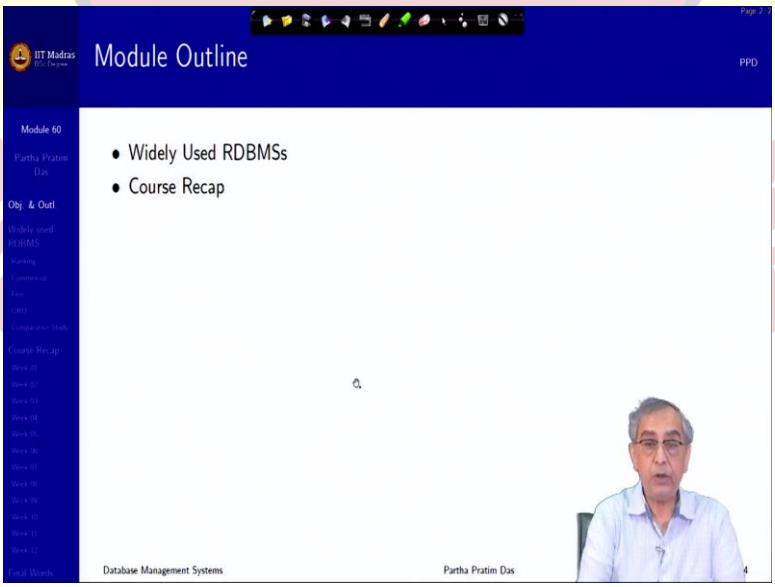

So, those are the two topics for this module.

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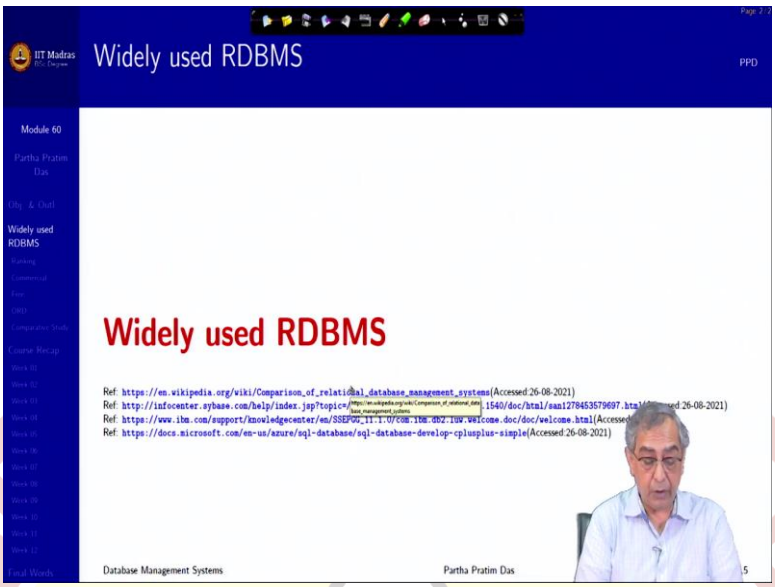

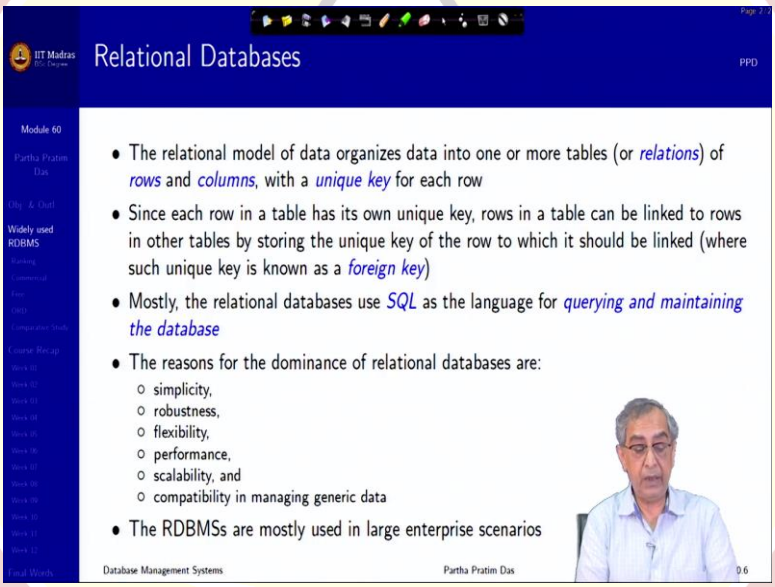

So, first we will talk about the widely used relational database management systems. The relational data model organizes data into one or more tables or relations as we have seen, which has rows and columns and unique key for each row. And since each row has a own unique key, the row of a table can be linked to rows of other tables using the unique key and we may have linking through the foreign key. We have learnt about SQL the language for querying and maintaining database which is mostly used by all of these relational databases.

And one of the reasons for dominance of relational databases, we will see the extent of this dominance soon are these properties like simplicity, robustness, flexibility, performance, scalability to a certain extent, to the extent of enterprise requirement, we have seen the issues

of scalability as well. And compatibility in managing generic data and RDBMS are mostly using large enterprise scenarios.

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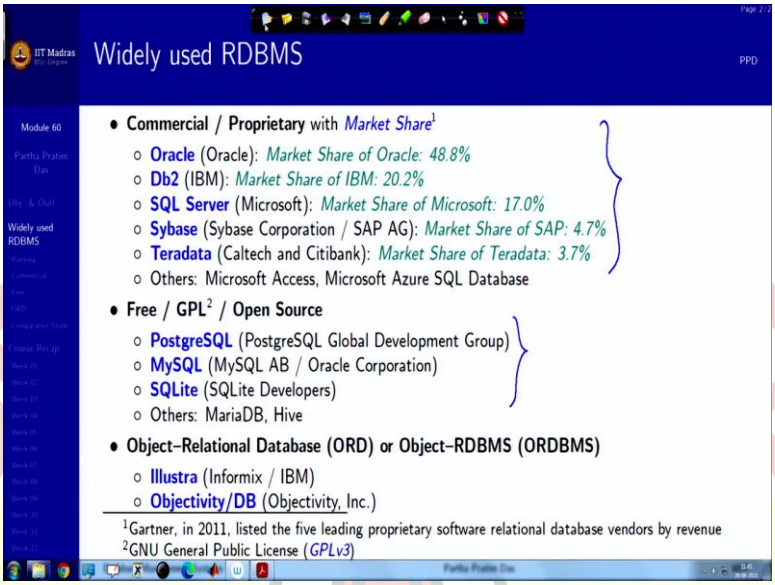

So, here, I present some very specific business data corresponding to the widely used RDBMSs they fall into broadly two categories commercial or proprietary where you will have to pay for using that database and the other is free or open source databases, which work under the GNU, General Public License and you may not need to pay primarily for using that database may have to pay for some maintenance.

Now, in this what is the way this is ranked is these are the top five relational databases according to the market share data. This is not absolutely recent data, this is as of 10 years back, but that is what comprehensively available. You can see that the Oracle from Oracle is dominating the market with about 48.8 percent of the revenue share followed by not closely but the next second next is Db2 from IBM with about 20 percent, SQL Server of Microsoft is close by with 17 percent and then much lower on the rank is Sybase which is primarily SAP 4.7 percent and Teradata and there are several others, which are also quite widely used and popular.

In terms of the free space, PostgreSQL, MySQL and SQLite are the three major dominating relational databases and there are others like MariaDB, Hive, Hive is primarily used for data warehousing. And there are some which are called object relational databases, which are relational databases having certain object linking properties like Illustra and objective DB this of course, are get merged into the NOSQL databases very soon.

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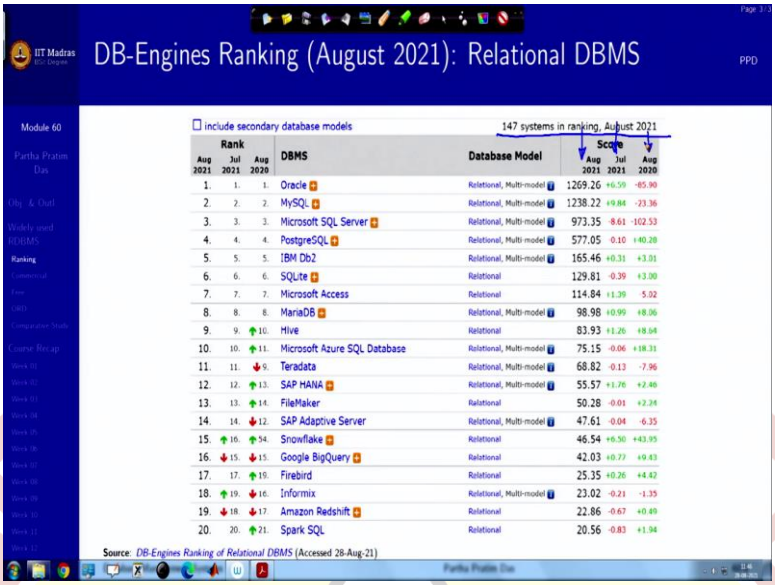

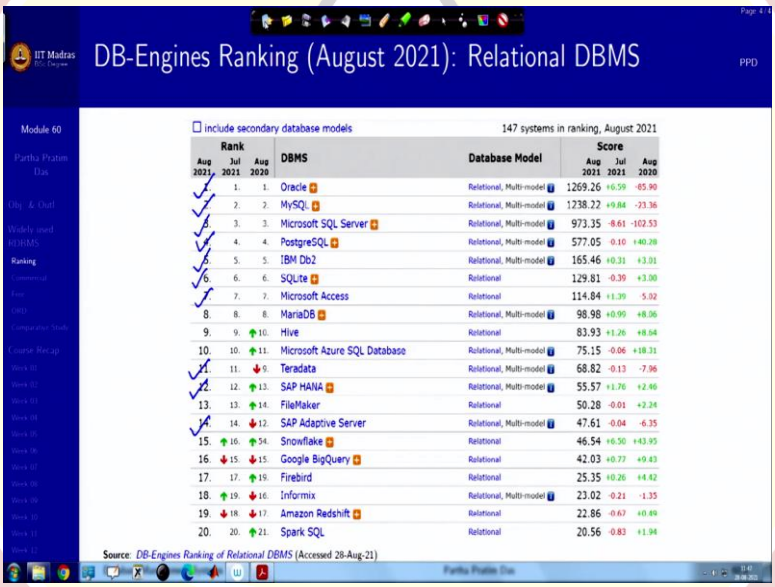

So, if you look into DB engines is a, is an organization which keeps on ranking the database servers or the database relational databases, all databases actually on a regular basis on a monthly basis. So, this is the latest ranking report where 147 database RDBMSs have been ranked as of August 2021. And these are this is kind of the rank score to understand the rank score you will have to study go to their site and study, the site link is given below and study that their actual approach of doing the analysis.

And on the next two columns, this show compared to the previous month, what is the increase or decrease increase as shown in green and decrease in red. And compared to the one year back, what is their increase or decrease. So, what you can see is Oracle and MySQL,

Oracle from proprietary and MySQL from the kind of public space are kind of the leaders when they are very closed by in this.

Similarly, next year followed our Microsoft SQL Server and PostgreSQL, which are again proprietary and free, they are in the next two slots followed by IBM Db, SQLite, your Microsoft Access, you have Teradata here, you have SAP HANA, SAP adaptive servers, these are actually which subsumes the Sybase databases. So, this is kind of the current ranking. So you can know that what do you what would be your expectation to see the different databases as you go for work.

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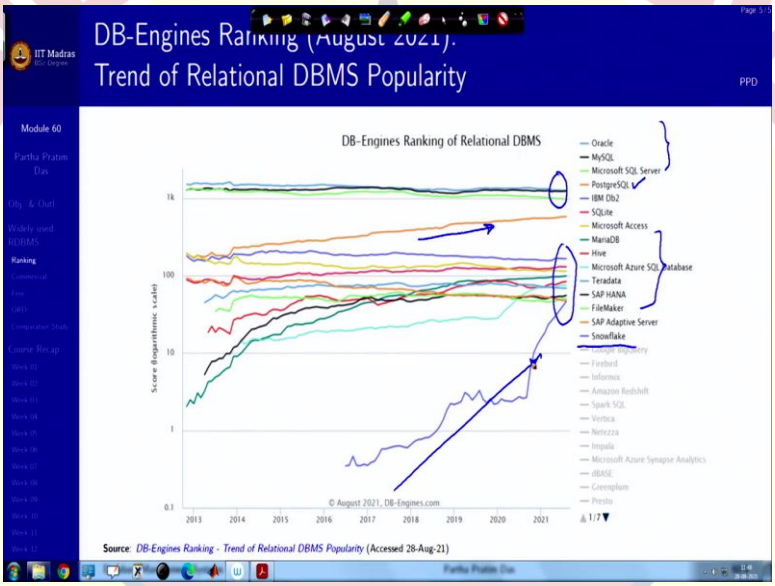

Now, this is a kind of a trend curve over the last about a decade from 2013. And here, you can see that how in terms of this is a y axis is a logarithmic scale to keep things because suddenly things are exponentially increasing, because the data is exponentially increasing. So, in the log scale, if something is flat, then it shows that that is the level of exponential increase and so on.

So, you can see that on the top are a couple of clattered are some of the three databases, RDBMSs which have kind of held on to the top position over a decade. So that is kind of the trend. Whilst, PostgreSQL has been increasing. So, it is getting more and more popular, you have then then you have a clutter of a number of different databases, which are kind of they started at different levels and over the last couple of years have kind of settled down at a similar kind of growth happening across them, those will be all of these databases there.

And you can see some of the recent databases like Snowflake, which started as a about 5 years back has had a very steep growth in terms of this. So, it is, I mean it is something to watch out for. So, this is kind of the trend overall.

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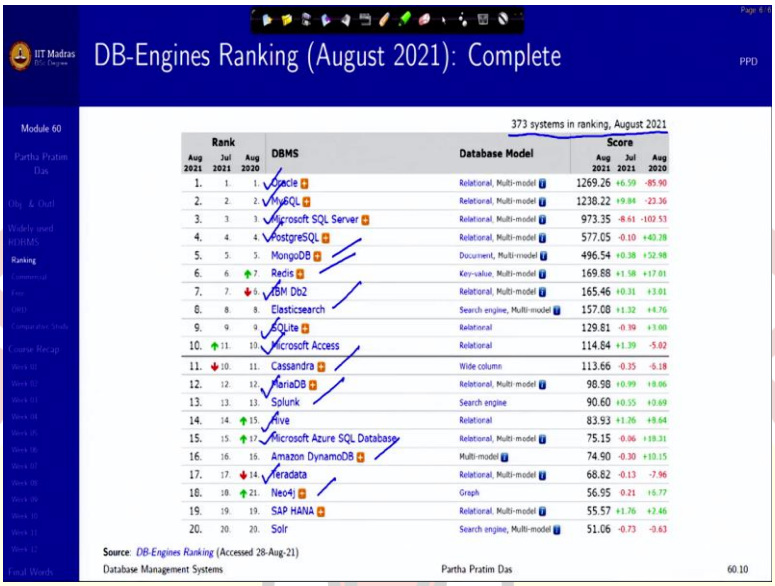

Now, I will also present you a data on what is happening in the entire you know market space of databases in general not only the relational, relational as well as non-relational including the NOSQL databases. So, if you take a look into that. So, it is this is a data of 377 systems, we are just presenting the top 20 the site has entire list as of the current month, August 2021.

And you can see that the top quite a few Oracle, MySQL, SQL Server PostgreSQL, the top four are still the relational databases. So, you can see that dominance that still goes on MongoDB is the, Redis these are the top non-relational or NOSQL databases which are dominating in the popularity space. IBM Db is again, relational Elasticsearch is little bit of a different it is, it is more a search engine than a typical database, but it has all the functionalities that the database needs, but the primary focus is on, on searching.

So that is also non-relational and has is a multi-modal database which has a lot of popularity. Again followed by relational databases SQLite Microsoft Access, followed by NOSQL Cassandra, relational MariaDB, Splunk search engine, relational Hive, relational Azure NOSQL Dynamo, relational Teradata, NOSQL Neo4j and so on.

So, you can see that actually the scalability and functionality decides a lot of things, but the space is very uniformly it is not, it is clearly that to a certain extent the relational database systems do dominate the space, but non-relational are catching up, the NOSQL is catching up

they have a, but typically what happens is for relational there are a large number of enterprise which use that because they need all those transactional online transaction processing OLTP services.

In terms of NOSQL, the each and every organization using it use for a very large data space very large user base like Facebook, like Amazon, like Google and so on. But, the number of such organizations number of such requirements are certainly less because there cannot be that huge organizations or applications in tens of thousands or hundreds of thousands. So, that is the kind of the differentiator.

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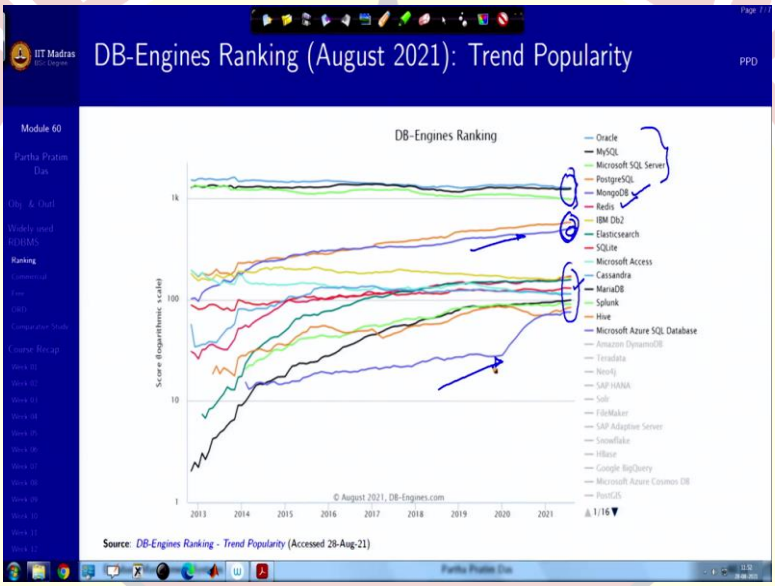

And if you look at the trend data of all databases, again, you will observe interesting things like on the top, the clutter is with the relational databases followed by MongoDB here, this is the one and Redis somewhere below that here. So, this is again, again, the kind of view again, see the three groups that we are seeing in the relational in terms of the trend of changes, but you can see that more or less most of the players are kind of in a stable position, except for again, PostgreSQL we saw is growing so is MongoDB.

So, these are these are the growing trends, naturally, this one has a which is Microsoft Azure SQL Database, which is also having a recent growing trend and so on. So, just keep on tracking even know what is happening in this space and what would be the appropriate things to learn further.

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Now in the next couple of slides, what are present and I am not going to discuss through them except for one or two. I present the basic information about these top proprietary as well as free databases say for Oracle, it is a database which kind of came into strong commercial existence in 1977. The latest version as of August 2021 is Oracle Database 19c, but different organizations would be using different versions.

And these are the application domains almost everything except for the non NOSQL kind of requirements. But even then, in some of those cases, there is backbone from Oracle. So primarily online transaction processing, data warehousing, and mixed applications are the primary. And Oracle specifically offers different services based on the Oracle database, which are sector specific. So for example, if you want a customer management, customer experience tracking kind of database to be built, you do not have to build it from scratch, Oracle customer experience gives you modules which can be directly customized for your requirement and used.

Languages all across the SQL are kind of, all across the relational databases are SQL or the procedural versions of that. There are further tools of many databases provide for facilitating development and connectivity as you have learned is typically I have tried to give some, some specifics but it they are typically in terms of I should say four buckets one is ODBC in general, which most databases support, then for Java you have JDBC, for Microsoft specific stuff, you will have the Microsoft dot net ADB ODB dot net, ADO etc. for Microsoft specific stuff.

And Python often has third party libraries provided in Python for doing this we have taken a big look into all that. So, I will not go through the details of everyone, you can read them study them and make a feeling about how the competitions are going on. And what you can get from different databases.

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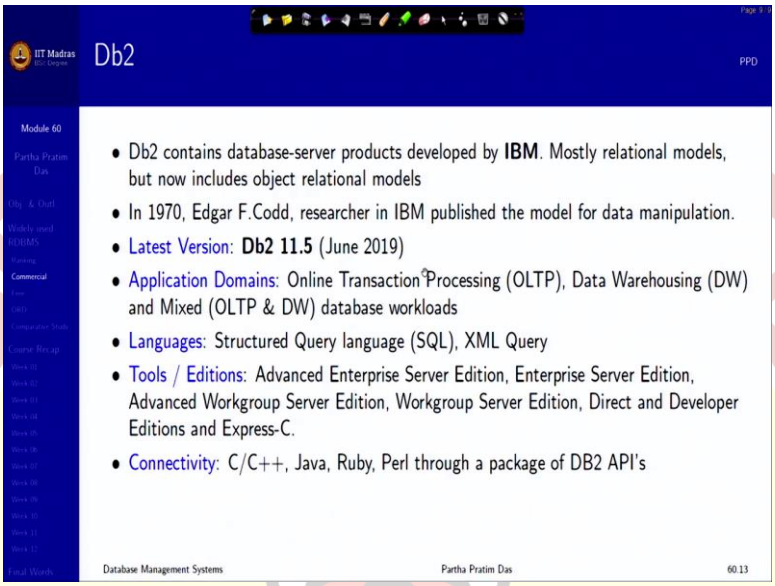

This is Db2 from IBM.

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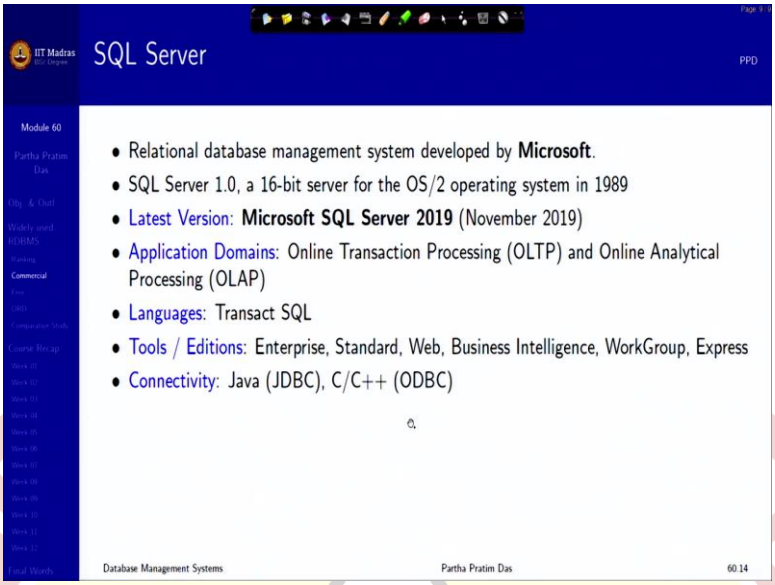

You have SQL Server from Microsoft, these are all competing as we have seen.

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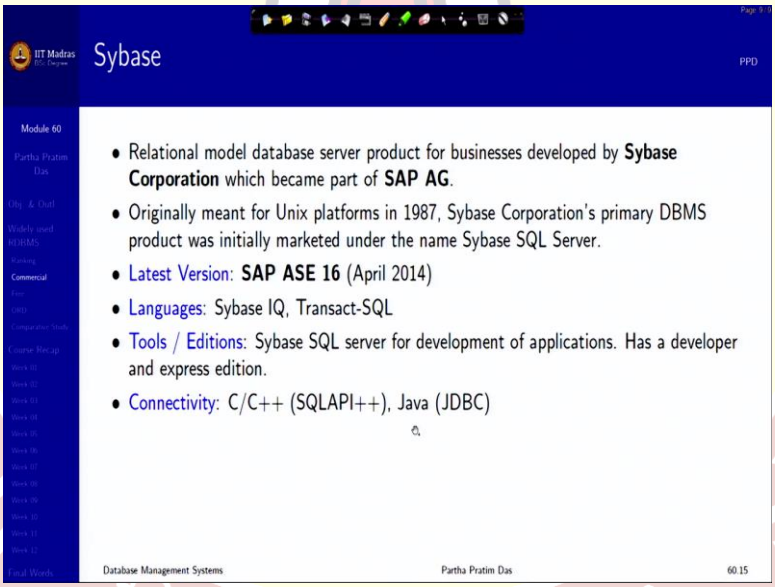

You have Sybase which used to be quite a competitor of Oracle in the late 80s and quite a much 90s, they eventually got acquired by SAP and they became the backbone of SAP. So, it mostly goes for SAP application transact SQL is a major component. So, they have specific SQL API's for use, which are built on ODBC.

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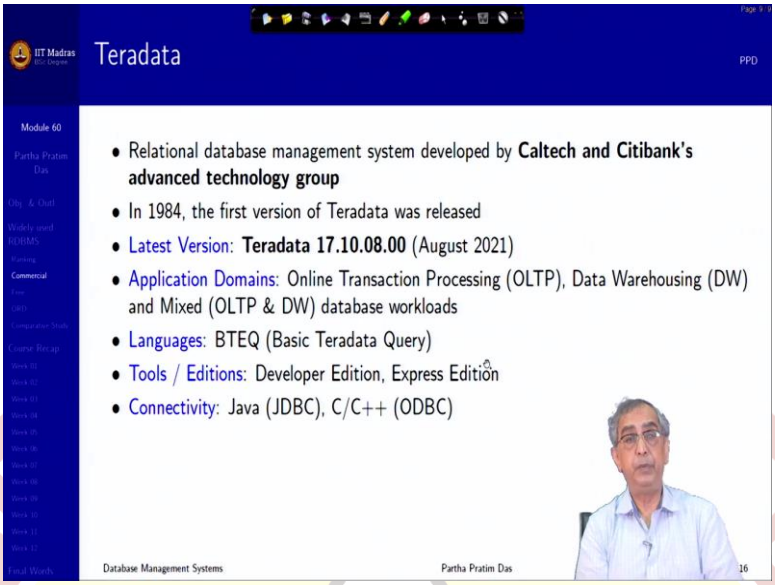

Teradata is also another which is, as you seen, quite popular.

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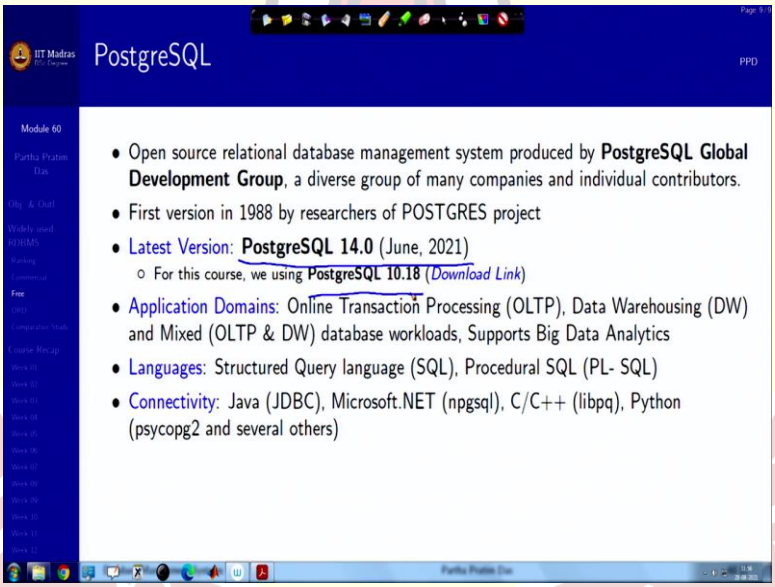

Now coming to the free and open source ones, PostgreSQL, MySQL are some of the leaders. Now this is the latest version of PostgreSQL as of June 2021, that is two months back, but we are using this in the course but we are using an old stable version, the download link is here in case you have not already downloaded it yet. And it can be used for several of the OLTP data warehousing and other applications. And you have, we have already discussed quite a bit about this particular database system.

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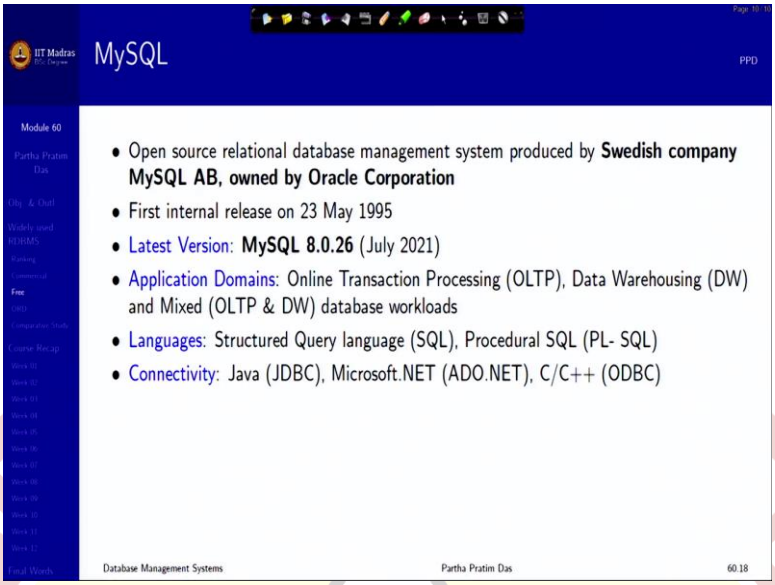

MySQL is another open database system, which was managed by the community but later, currently is owned by Oracle, but it is still open source and used, can be used freely.

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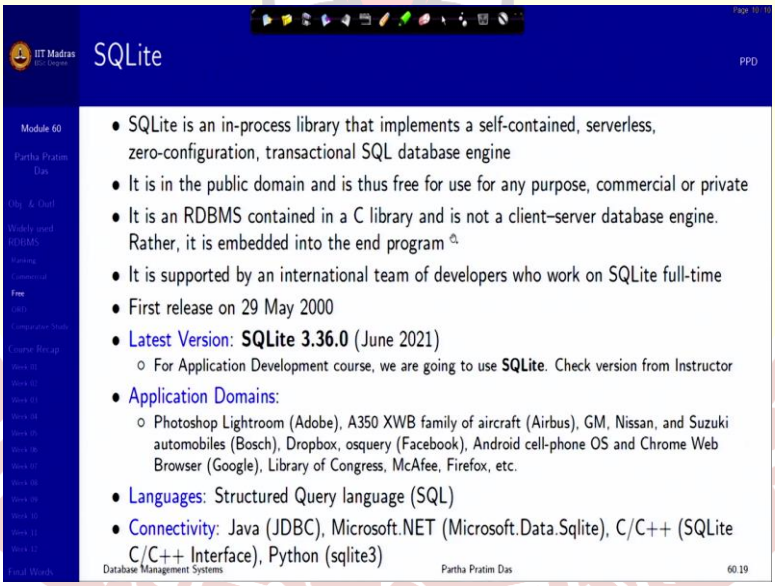

SQLite is a community initiative led by a group of international independent developers and its latest version is of June 2021. This is going to be the database system that your application development course which goes hand in hand with this DBMS course will use, you have to check the version from the instructor. And here have given certain, certain indicative big applications which are based on SQLite and they come from big companies like Adobe, Bosch, Google, Airbus, McAfee and so on.

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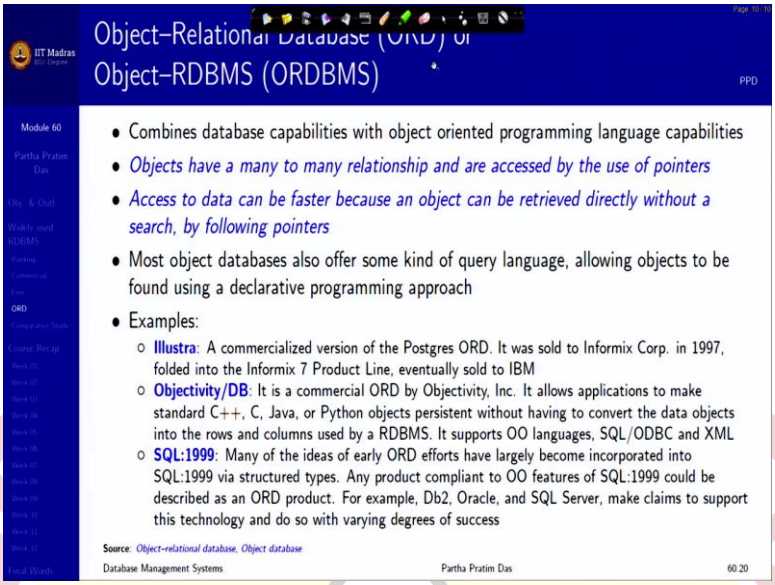

Now, there are object relational databases like Illustra and Object Db which have, can support many to many relationships accessed by the use of pointers. So, that gets rid of the excessive use of join, which is the most costly operation in a relational database. So, its advantages in that way, their penetration is still not very high, because strong object based databases have actually gone into the NOSQL space as we have already seen.

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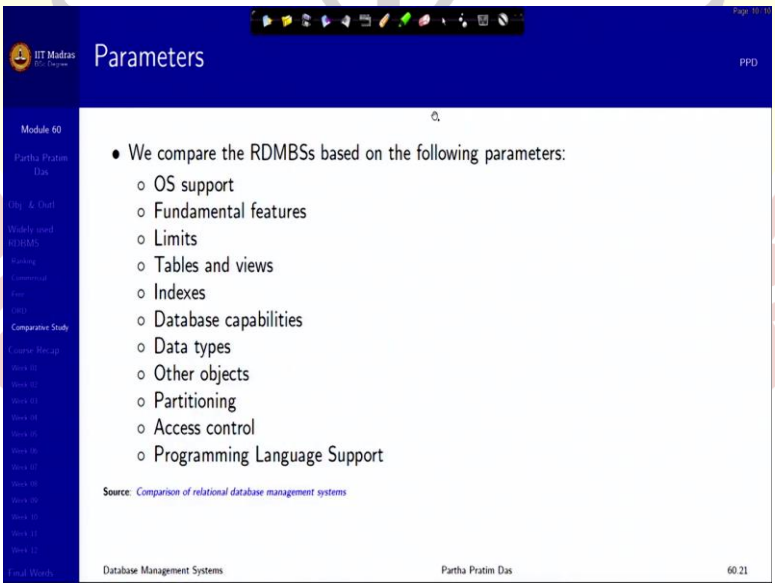

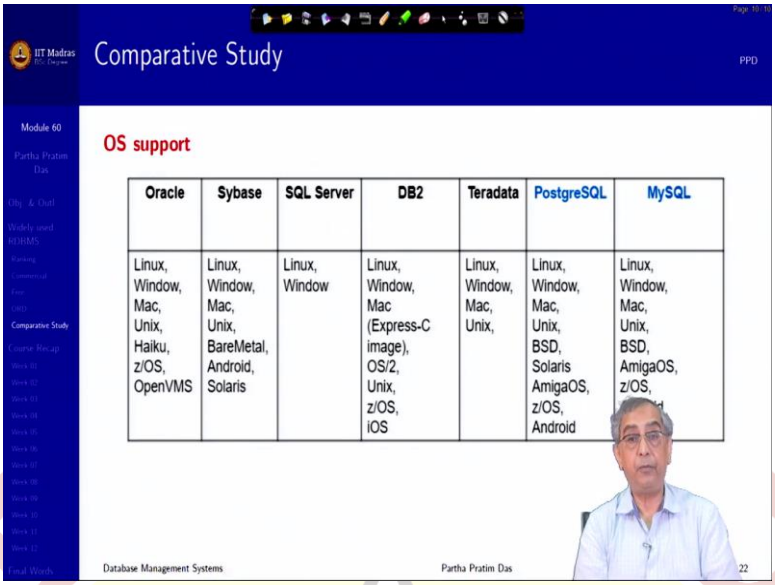

Oracle	Sybase	SQL Server	DB2	Teradata	PostgreSQL	MySQL
Linux, Window, Mac, Unix, Haiku, z/OS, OpenVMS	Linux, Window, Mac, Unix, BareMetal, Android, Solaris	Linux, Window	Linux, Window, Mac (Express-C image), OS/2, Unix, z/OS, iOS	Linux, Window, Mac, Unix,	Linux, Window, Mac, Unix, BSD, Solaris AmigaOS, z/OS, Android	Linux, Window, Mac, Unix, BSD, AmigaOS, z/OS,

Now, in the in the next couple of slides, you will see the comparison of these databases based on various parameters, how do they compare with in terms of operating system support fundamental features, limits, tables, views and so on. So again, I will not go through them in detail. For example, here we are showing the five, top five proprietary databases and the top three databases as to what and the operating system that they support.

So, Linux, Windows Mac kind of is universal, almost everybody supports but like SQL server does not support you can understand the reason they do not. So, but there are several other operating systems where which has support variedly like PostgreSQL is supported on Android, MySQL is supported on Android. Sybase is supported on Android. So, these are useful for your mobile application development.

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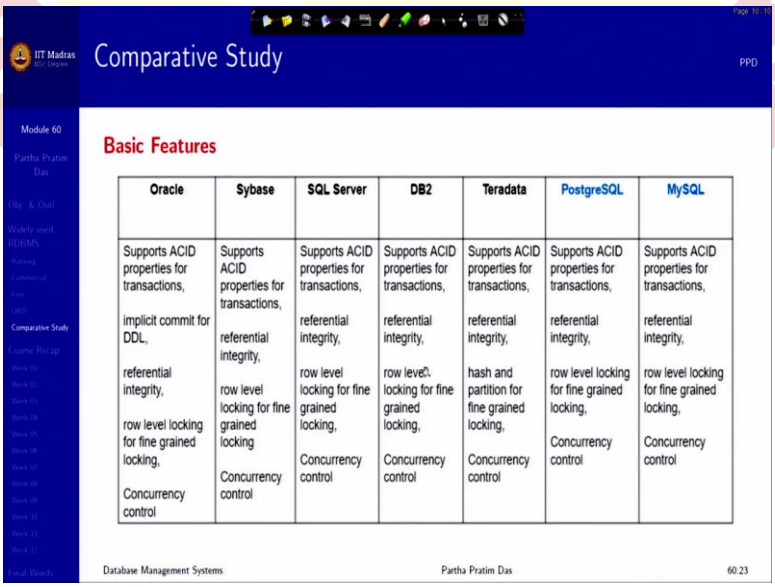

Oracle	Sybase	SQL Server	DB2	Teradata	PostgreSQL	MySQL
Supports ACID properties for transactions, implicit commit for DDL, referential integrity, row level locking for fine grained locking, Concurrency control	Supports ACID properties for transactions, referential integrity, row level locking for fine grained locking Concurrency control	Supports ACID properties for transactions, referential integrity, row level locking for fine grained locking, Concurrency control	Supports ACID properties for transactions, referential integrity, row level locking for fine grained locking, Concurrency control	Supports ACID properties for transactions, referential integrity, hash and partition for fine grained locking, Concurrency control	Supports ACID properties for transactions, referential integrity, row level locking for fine grained locking, Concurrency control	Supports ACID properties for transactions, referential integrity, row level locking for fine grained locking, Concurrency control

These are a list of basic features they support, most of the basic ones would be this would be supported by everyone the special things which will vary.

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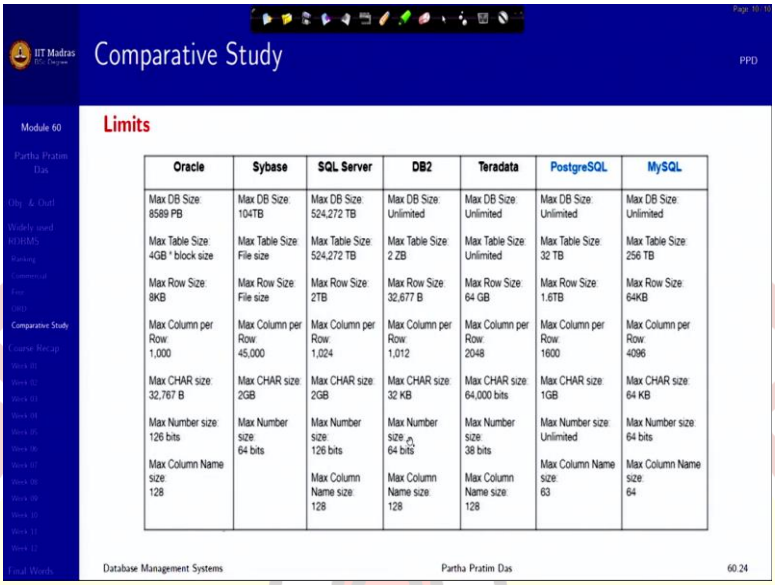

Oracle	Sybase	SQL Server	DB2	Teradata	PostgreSQL	MySQL
Max DB Size: 8589 PB	Max DB Size: 104TB	Max DB Size: 524,272 TB	Max DB Size: Unlimited	Max DB Size: Unlimited	Max DB Size: Unlimited	Max DB Size: Unlimited
Max Table Size: 4GB * block size	Max Table Size: File size	Max Table Size: 524,272 TB	Max Table Size: 2 ZB	Max Table Size: Unlimited	Max Table Size: 32 TB	Max Table Size: 256 TB
Max Row Size: 8KB	Max Row Size: File size	Max Row Size: 2TB	Max Row Size: 32,677 B	Max Row Size: 64 GB	Max Row Size: 1.6TB	Max Row Size: 64KB
Max Column per Row: 1,000	Max Column per Row: 45,000	Max Column per Row: 1,024	Max Column per Row: 1,012	Max Column per Row: 2048	Max Column per Row: 1600	Max Column per Row: 4096
Max CHAR size: 32,767 B	Max CHAR size: 2GB	Max CHAR size: 2GB	Max CHAR size: 32 KB	Max CHAR size: 64,000 bits	Max CHAR size: 1GB	Max CHAR size: 64 KB
Max Number size: 126 bits	Max Number size: 64 bits	Max Number size: 126 bits	Max Number size: 64 bits	Max Number size: 38 bits	Max Number size: Unlimited	Max Number size: 64 bits
Max Column Name size: 128		Max Column Name size: 128	Max Column Name size: 128	Max Column Name size: 128	Max Column Name size: 63	Max Column Name size: 64

There are different limits are set by different databases typically, so you can see that so for your growth, these factors are important to know.

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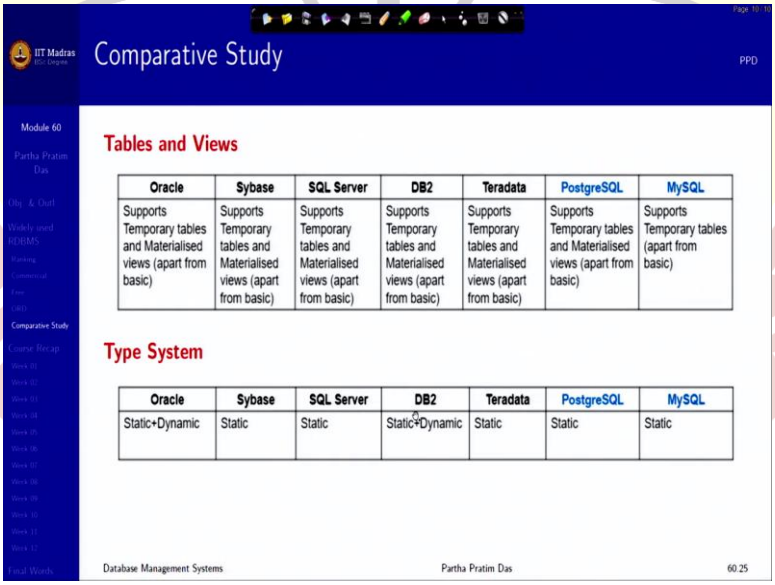

Oracle	Sybase	SQL Server	DB2	Teradata	PostgreSQL	MySQL
Supports Temporary tables and Materialised views (apart from basic)	Supports Temporary tables and Materialised views (apart from basic)	Supports Temporary tables and Materialised views (apart from basic)	Supports Temporary tables and Materialised views (apart from basic)	Supports Temporary tables and Materialised views (apart from basic)	Supports Temporary tables and Materialised views (apart from basic)	Supports Temporary tables (apart from basic)

Oracle	Sybase	SQL Server	DB2	Teradata	PostgreSQL	MySQL
Static+Dynamic	Static	Static	Static+Dynamic	Static	Static	Static

Tables and views type systems, they vary between type systems vary between being static, and being a combination of static and dynamic.

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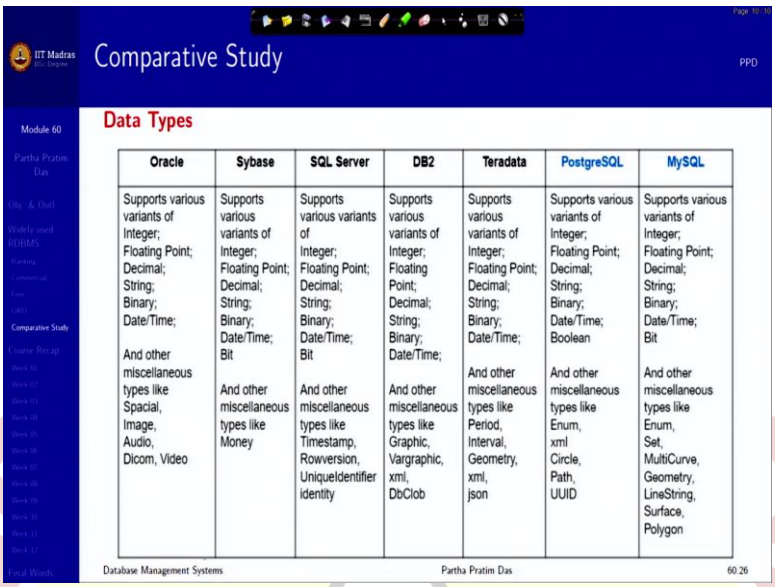

Oracle	Sybase	SQL Server	DB2	Teradata	PostgreSQL	MySQL
Supports various variants of Integer; Floating Point; Decimal; String; Binary; Date/Time; And other miscellaneous types like Spacial, Image, Audio, Dicom, Video	Supports various variants of Integer; Floating Point; Decimal; String; Binary; Date/Time; Bit And other miscellaneous types like Money	Supports various variants of Integer; Floating Point; Decimal; String; Binary; Date/Time; Bit And other miscellaneous types like Timestamp, Rowversion, UniqueIdentifier identity	Supports various variants of Integer; Floating Point; Decimal; String; Binary; Date/Time; And other miscellaneous types like Graphic, Vargraphic, xml, DbClob	Supports various variants of Integer; Floating Point; Decimal; String; Binary; Date/Time; And other miscellaneous types like Period, Interval, Geometry, xml, json	Supports various variants of Integer; Floating Point; Decimal; String; Binary; Date/Time; Boolean And other miscellaneous types like Enum, xml Circle, Path, UUID	Supports various variants of Integer; Floating Point; Decimal; String; Binary; Date/Time; Bit And other miscellaneous types like Enum, Set, MultiCurve, Geometry, LineString, Surface, Polygon

So, you certainly the expectation is not that he will learn these tables by heart, but these are more like references to give you a feel of the variability you have and the commonality of these properties that you have already learned in the course.

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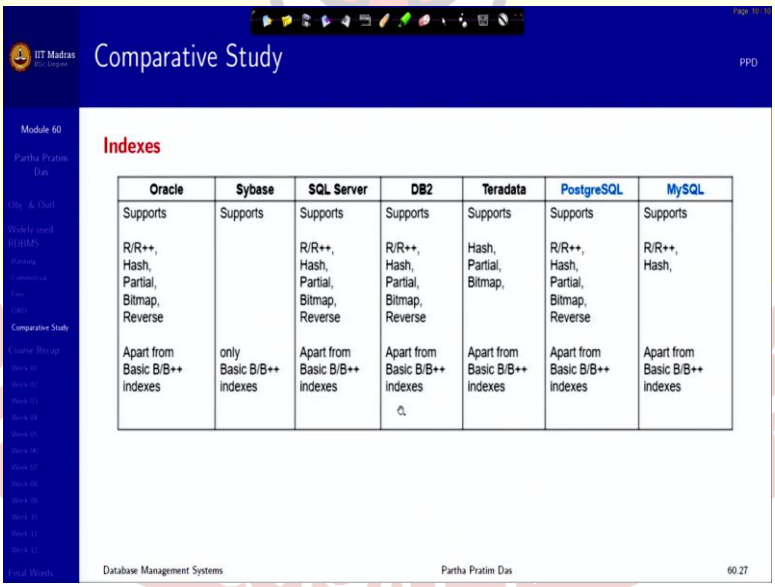

Oracle	Sybase	SQL Server	DB2	Teradata	PostgreSQL	MySQL
Supports R/R++, Hash, Partial, Bitmap, Reverse Apart from Basic B/B++ indexes	Supports only Basic B/B++ indexes	Supports R/R++, Hash, Partial, Bitmap, Reverse Apart from Basic B/B++ indexes	Supports R/R++, Hash, Partial, Bitmap, Reverse Apart from Basic B/B++ indexes	Supports Hash, Partial, Bitmap, Apart from Basic B/B++ indexes	Supports R/R++, Hash, Partial, Bitmap, Reverse Apart from Basic B/B++ indexes	Supports R/R++, Hash, Apart from Basic B/B++ indexes

So, you should be able to get into any of these databases and seamlessly start working. Of course, you will have to defer to the specific manual for all this information.

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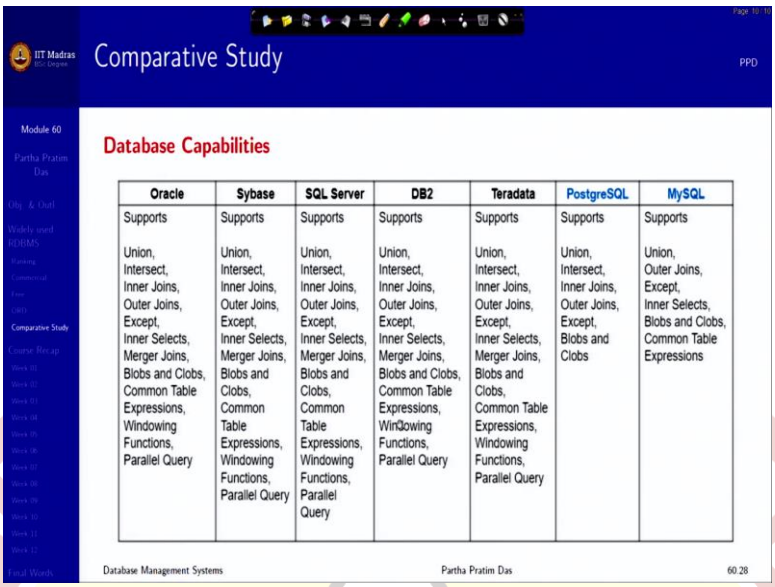
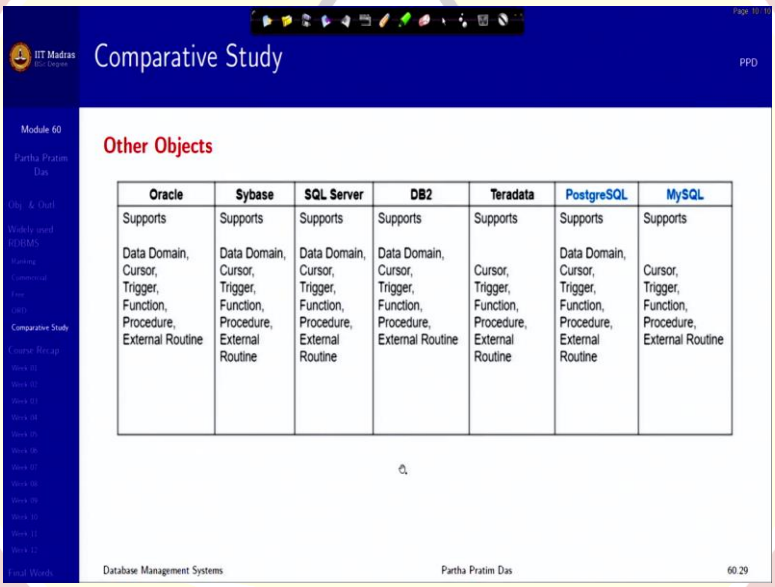


Oracle	Sybase	SQL Server	DB2	Teradata	PostgreSQL	MySQL
Supports Union, Intersect, Inner Joins, Outer Joins, Except, Inner Selects, Merger Joins, Blobs and Clobs, Common Table Expressions, Windowing Functions, Parallel Query	Supports Union, Intersect, Inner Joins, Outer Joins, Except, Inner Selects, Merger Joins, Blobs and Clobs, Common Table Expressions, Windowing Functions, Parallel Query	Supports Union, Intersect, Inner Joins, Outer Joins, Except, Inner Selects, Merger Joins, Blobs and Clobs, Common Table Expressions, Windowing Functions, Parallel Query	Supports Union, Intersect, Inner Joins, Outer Joins, Except, Inner Selects, Merger Joins, Blobs and Clobs, Common Table Expressions, Windowing Functions, Parallel Query	Supports Union, Intersect, Inner Joins, Outer Joins, Except, Inner Selects, Merger Joins, Blobs and Clobs, Common Table Expressions, Windowing Functions, Parallel Query	Supports Union, Intersect, Inner Joins, Outer Joins, Except, Blobs and Clobs	Supports Union, Outer Joins, Except, Inner Selects, Blobs and Clobs, Common Table Expressions



Oracle	Sybase	SQL Server	DB2	Teradata	PostgreSQL	MySQL
Supports Data Domain, Cursor, Trigger, Function, Procedure, External Routine	Supports Data Domain, Cursor, Trigger, Function, Procedure, External Routine	Supports Data Domain, Cursor, Trigger, Function, Procedure, External Routine	Supports Data Domain, Cursor, Trigger, Function, Procedure, External Routine	Supports Cursor, Trigger, Function, Procedure, External Routine	Supports Data Domain, Cursor, Trigger, Function, Procedure, External Routine	Supports Cursor, Trigger, Function, Procedure, External Routine

Indexes, database capabilities. Other kinds of objects, besides just the data tables.

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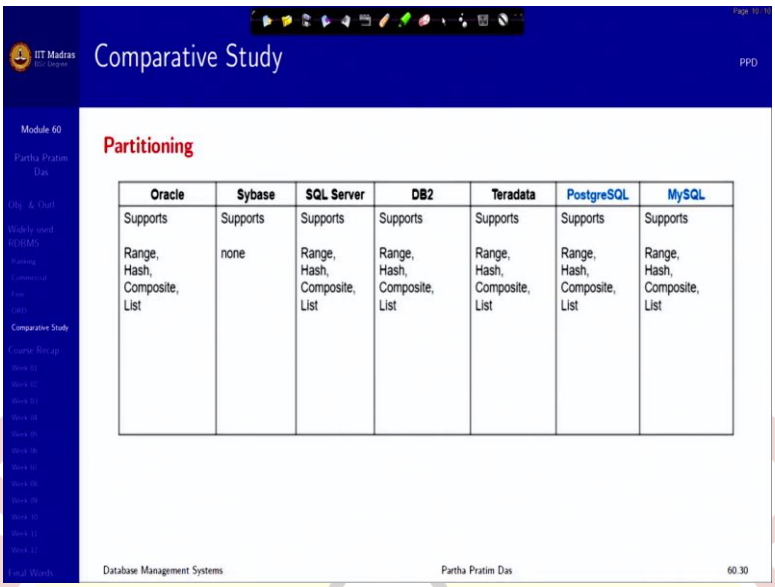

Oracle	Sybase	SQL Server	DB2	Teradata	PostgreSQL	MySQL
Supports Range, Hash, Composite, List	Supports none	Supports Range, Hash, Composite, List	Supports Range, Hash, Composite, List	Supports Range, Hash, Composite, List	Supports Range, Hash, Composite, List	Supports Range, Hash, Composite, List

How much it can be partitioned, some databases can be more partitioned, some databases cannot be partitioned, those kinds of information.

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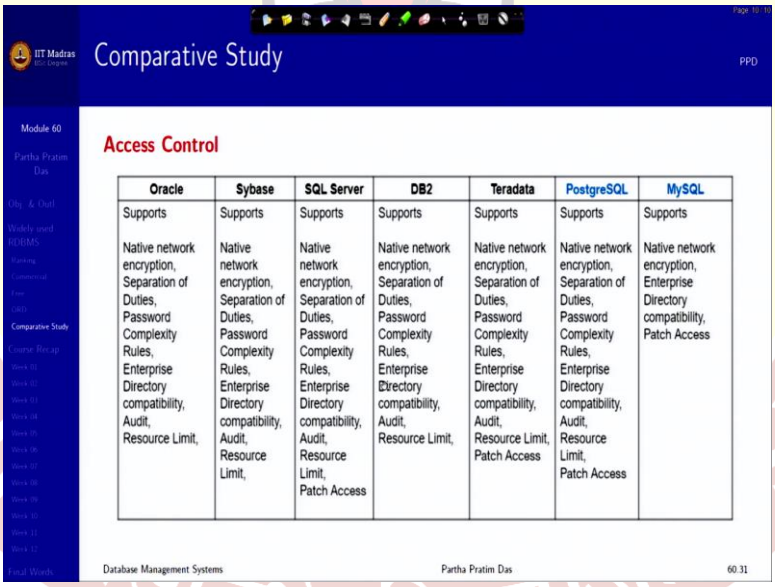

Oracle	Sybase	SQL Server	DB2	Teradata	PostgreSQL	MySQL
Supports Native network encryption, Separation of Duties, Password Complexity Rules, Enterprise Directory compatibility, Audit, Resource Limit,	Supports Native network encryption, Separation of Duties, Password Complexity Rules, Enterprise Directory compatibility, Audit, Resource Limit,	Supports Native network encryption, Separation of Duties, Password Complexity Rules, Enterprise Directory compatibility, Audit, Resource Limit, Patch Access	Supports Native network encryption, Separation of Duties, Password Complexity Rules, Enterprise Directory compatibility, Audit, Resource Limit,	Supports Native network encryption, Separation of Duties, Password Complexity Rules, Enterprise Directory compatibility, Audit, Resource Limit, Patch Access	Supports Native network encryption, Separation of Duties, Password Complexity Rules, Enterprise Directory compatibility, Audit, Resource Limit, Patch Access	Supports Native network encryption, Enterprise Directory compatibility, Patch Access

The access control information, the authentication facilities available in different database systems and so on. So, these are this is just to give you whole idea of the space which is complex, growing full of opportunity for you to jump in and make your lifelong career in this because databases are something that are not going to disappear from life.

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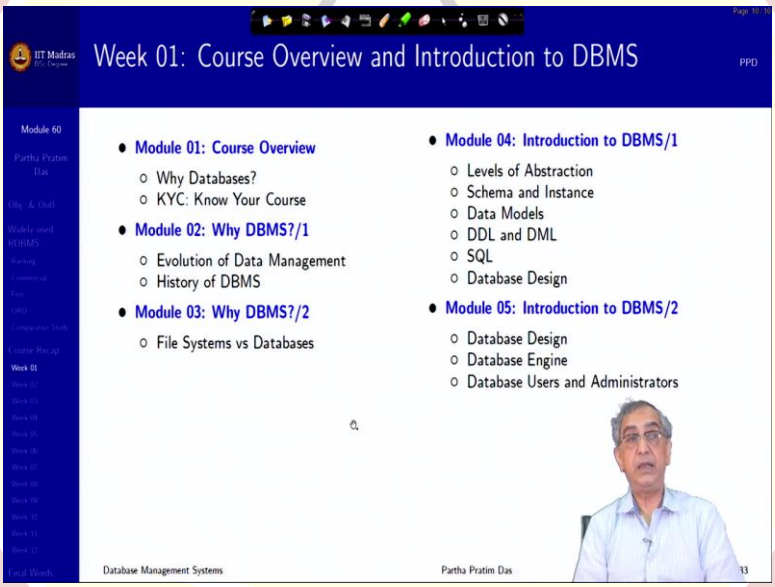

Now, coming to the course recap, this is just a overall summary of what we did week wise and in each one of them, I have mentioned the key points from the modules, 5 modules that every week had. So, in week 1, we had a course overview and we introduced the basic concept of the data DBMS why we need it as against the file system and what are the typical database design and engine considerations.

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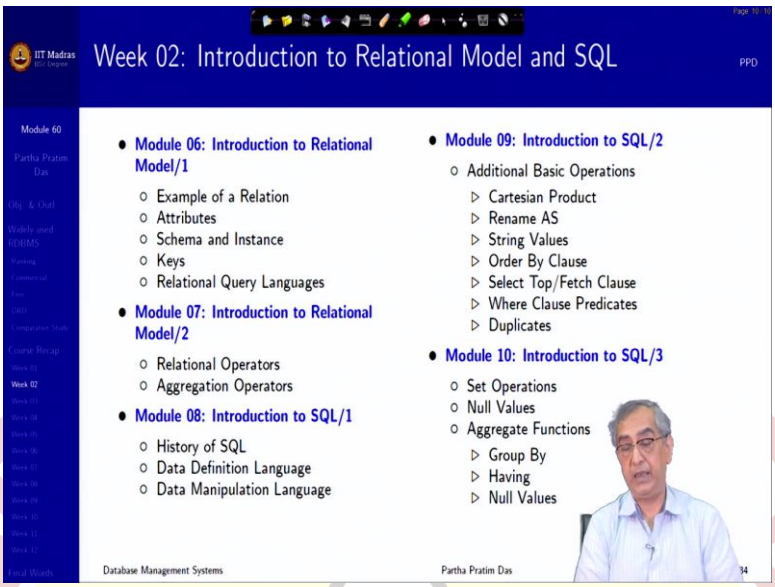

We followed it up by introducing the relational model and SQL. So, kind of relational model and SQL are kind of synonymous. So, that is the reason you can you can see that non-relational is being referred to as NOSQL, not no relational. So, we talked about the relational model the basic features of that and introduced you to the basic formulation of relational query which starts with data definition and manipulation languages and has additional features.

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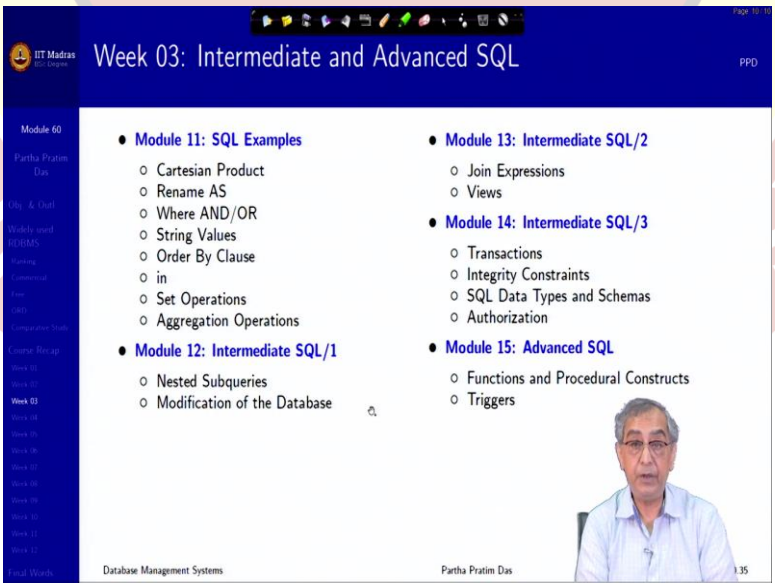

We moved on to delve deeper into the relational at intermediate level taking examples, talking about nested queries, joint expression views, transactions in SQL integrities, authorization in SQL and took up some advanced SQL features also I would not say that we

have comprehensively covered everything, but we have tried to be solid on the basic and intermediate side and somewhat sketchy on the advanced side because you can always want to know the fundamentals you can always read up and move forward in SQL.

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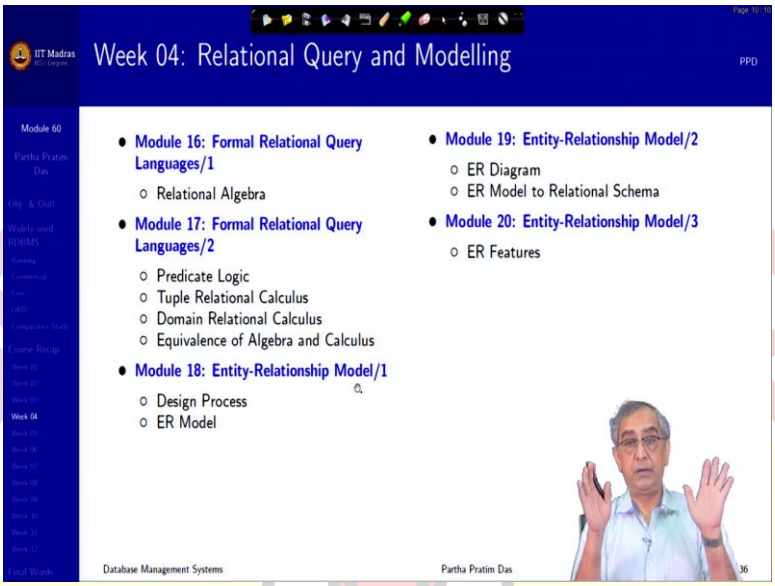

Being a database expert is necessarily whichever databases you work with at least expertise with SQL is required even for non NOSQL I have mentioned if you, if you recall that there are SQL (())(22:06) which are widely used in a good range of NOSQL databases. Then we moved on to relational query models and modeling of the real world in general.

So, we talked about relational algebra, calculi with certain introduction to predicate logic which you must brush up, the discrete structures you must brush up, talked about their equivalence and then moved on to modeling using the ER models, ER diagram conversion of ER models to relational schema, Entity Relationship features and so on.

And I mentioned that these are kind of restricted subset of a generalized modeling that you have today, which is called UML, Universal Modeling Language which we did not have time here to to study in depth because it talks about several other things, but if you are specializing from relational to NOSQL, I would strongly suggest that you also study UML in depth.

Because of the simple reason that ER models are very restrictive and more suitable for just relational modeling, the more of the connective modeling, more of the hierarchical modeling more of the inter connection, the sequence modeling, activity modeling you need which are very typical of more complex applications will be available in UML on top of what you have in the ER model.

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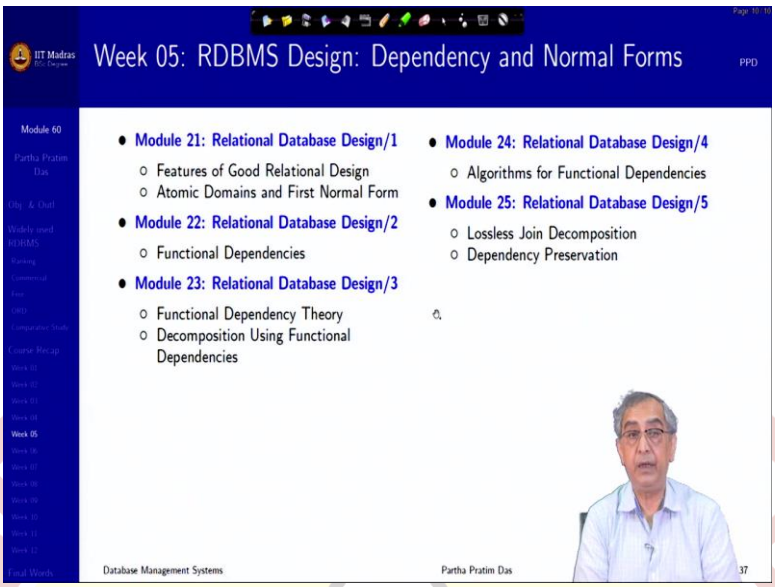

Moving on, we dealt really deep into the relational database design which is a, which is a very formalized subject. And we looked at how the mathematically the real world constraints can be modeled, what is atomic domains and normal forms and why are normal forms important, starting from the first normal form going upwards and then how the constraints of the real world can be modeled in terms of dependencies.

And we looked at the functional dependency theory, the first level of dependency theory, the and the decomposition using the functional dependencies, the algorithms for that, your lossless joint, dependency preservation, what are the properties you need.

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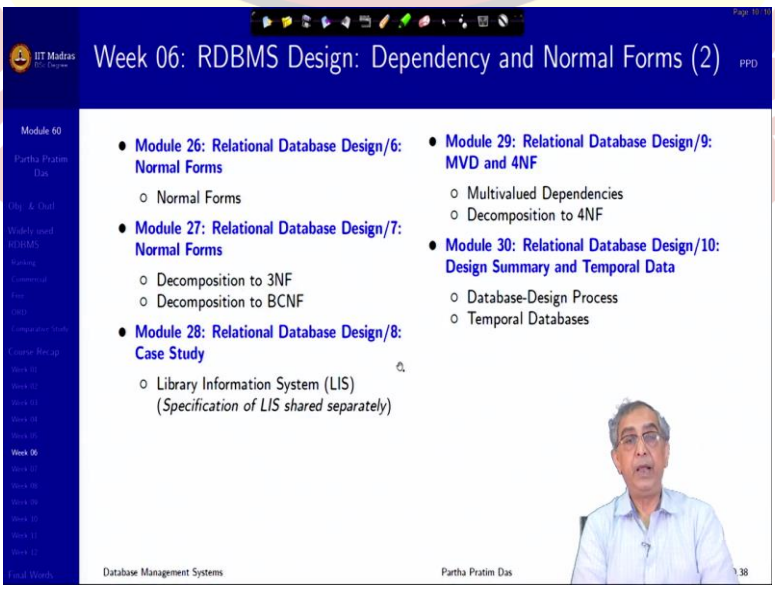

And we carried on into the sixth week in terms of talking more about the normal forms going to the second normal form, decomposition into third normal form, decomposition into Boyce Code normal form. Took an extensive example to illustrate the basic design refinement process using these normal forms and functional dependency constraints. And remember that along with this module 28 you will also need a specification document of the LIS which will be shared separately.

And moving on we took a good look in that relational database design in terms of multivalued dependency when it attributes necessarily have multiple values, which is also very common, and how do you decompose into 4NF which is more powerful than the BCNF as well.

And finally, we talked about the database design process and touched upon the issues of handling modeling time in a database. This is, this again becomes a very, very big question in terms of NOSQL databases, but relational databases can also model temporal data to a good extent, time series data, spatial data all of these can be modeled in relational as well.

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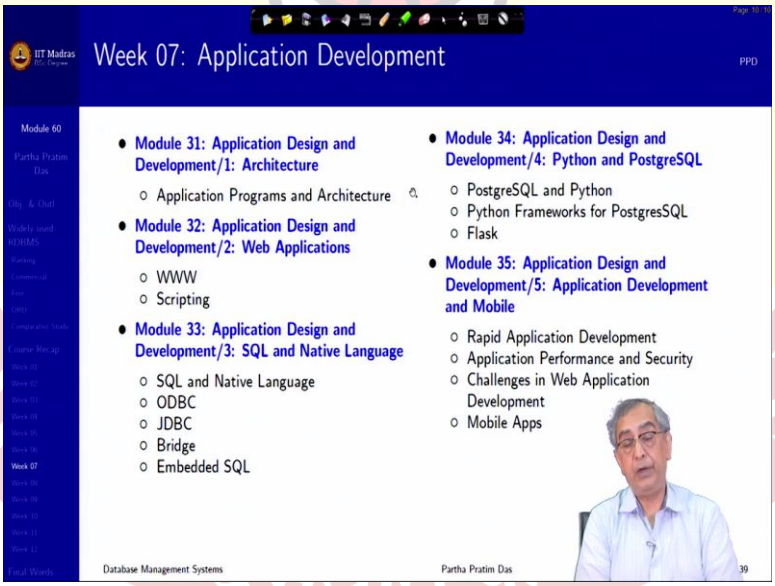

Week 7 is what we will, where we discuss that how to develop applications based on the databases that you have already created. So, or you are intending to create, so what does it take to develop an application, how to architect the application in terms of two tier, three tier or n tier, what is typically the web application requirements, the mobile application requirements and so on.

How you can use free databases like PostgreSQL, Python or frameworks like flask in terms of doing that, what is the role of rapid application development in this, how do you connect easily from a language, from a programming language to a database system to write the intermediate layer of the application and so on.

This is something which you will find will have quite some overlap with your application development course, many of these ideas will be re-iterated, reinforced in your application development course and even I will emphasize that you really need to learn that course while along with the database, so that you can become a powerful software engineer in the database application space.

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Next we took a having done the basic modeling and design, we took into the other aspects of database, more of the physical and temporal aspects, how do you manage storage, initial modules on this talk in general about background of data structures, which are primarily in memory and then we moved on to talking about the different physical storage and physical organization of the file organization of the data, comprehensively trying to look at what are the options available to store your files, your data files, your index files and so on.

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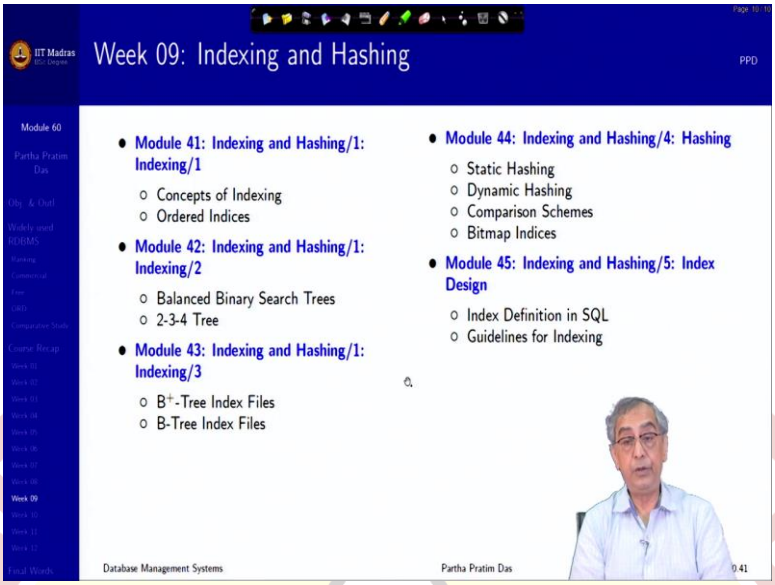

This naturally led to the design and considerations for indexing because indexing as we have learned is you can organize the data in only one way, based on the key primary key or something, but if you have the requirement of searching and updating the data in multiple, through multiple different attributes, then you need to have easy access through those and that can be done in terms of creating indexes. We again went back to the in-memory data structures for that what is balanced binary, search trees quick review of that.

And went on to discussing the 2-3-4 tree which is a very unique tree, where the height always remains the same for all the leaf nodes and you have flexible nodes having multiple number of keys going in them. And then we could generalize it in terms of the B plus tree, which are widely used for not only for index files, but actually also for the data files and talked about the specialization of B trees.

We looked at the, in the same week we looked at the other option for indexing which is through hashing, which is kind of is a mathematical mapping from a, from an attribute value to a specific address location where the particular record can be found.

And we did, basics of static dynamic hashing compared the schemes and talked about bitmap indices and also looked at actually in SQL how can you create indexes and give your instruction that this is where I need a bitmap index, this is what I need a composite index, how do I do a composite index and so on. So, this will be very required for your query formulation and optimization.

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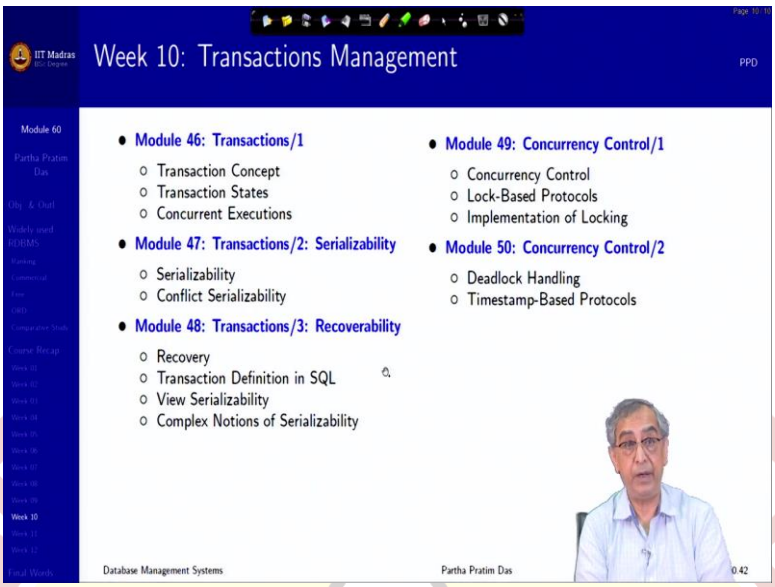

The core of the whole thing revolves around transaction the tasks that you want to do. So, we introduced the concept of transactions, states, concurrency of that and obviously the moment you make transactions concurrent, then the possibility of having incorrect result comes up. So, we talked about serializability, the conflict serializability we talked about the recoverability of transactions if you have to roll them back. If you have to, you have some failure and you cannot complete a transaction, we talked about the view serializability in different ranges of that.

And saw that well with all these, we finally end up may end up with a very few serial serializable schedules, which are polynomial time decidable, because view serializabilities and be complete.

So, we talk about the different mechanisms that are available to get the concurrency control on the fly, keeping the core idea of isolation in view that a particular data item can be read by two or more transactions at the same time, but can be written by only one transaction. And during that time, it has to be exclusive. So, we talked about log based protocols, two phase locking, time based protocol and consequences of that in terms of deadlock, live log and how to handle those.

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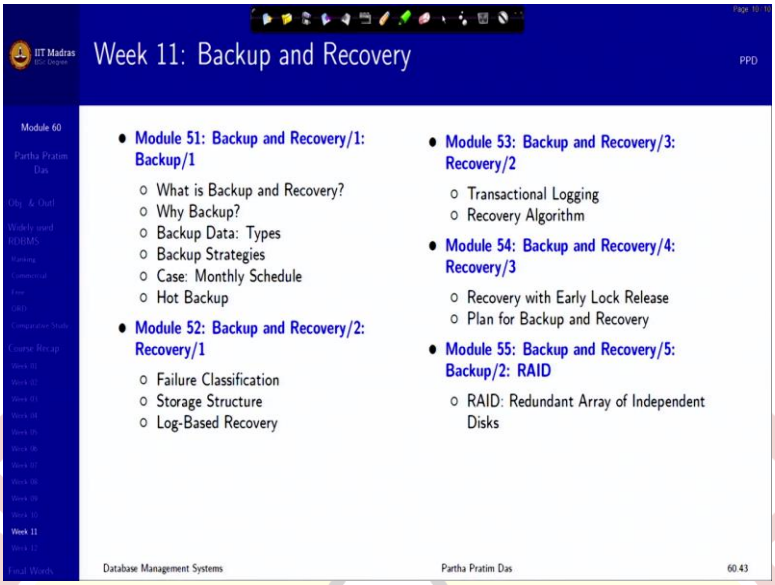

Week 11 focus primarily on backup and recovery, which is very required for the persistency of the database. What are the backup strategies, the backup plan and what are the ways that you can recover from that by using particularly the transaction logging the log based recovery, which is the key idea that we introduced here and revolved our strategies around that and we ended with a kind of a stable storage description of redundant array of independent disks.

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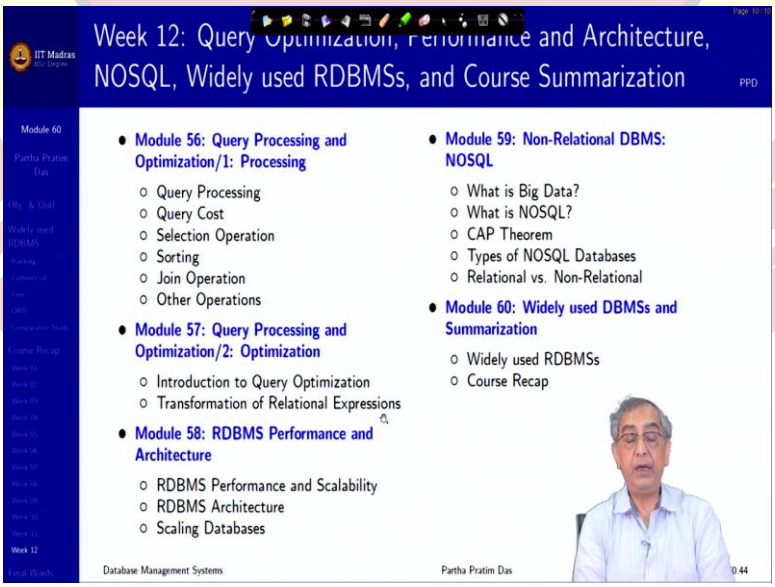

And in the current week, we started with query processing and optimization making evaluation plans, which are good in terms of estimated cost and how to use that effectively, we talked about. Then we digressed a little bit in the last two modules talking about the

performance of RDBMS, how does the architecture affect that performance and how can you scale the, scale the infrastructure, scale the RDBMS along with the scaling of the application.

And we saw that horizontal scaling scale out is the primary way to go and in view of that, we took a look into what is emerging in terms of big data, why do we have not only SQL that is NOSQL in place, and what are the compromise that you do in terms of so the CAP requirements that is Consistency, Availability and Partitioning all of the three you cannot get together.

So, you have to decide which two primarily you want and there are different protocols like Gossip protocol and so on, which allow you to compromise in a restrictive or in a controlled way. And in this module, as it is you have seen we have talked about the widely use RDBMSs.

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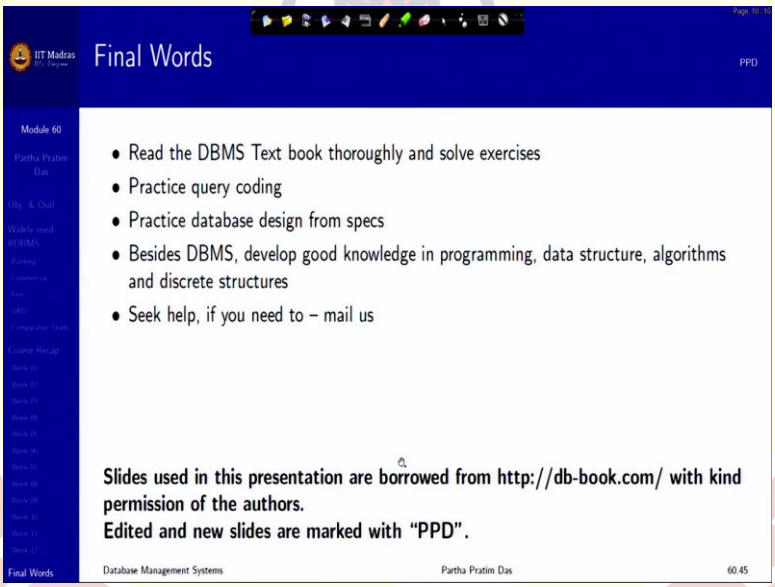

So, it is it is my pleasure that I could do work with you, work through this particular course, which is very close to my heart, which I think is a very effective technology to learn, to grow in the industry as a software developer, as a database engineer, as a data systems engineer and growing to the DBAs and data scientists and so on.

So, to prepare well for this course read the textbook very thoroughly, solve the exercise, practice query coding, this is very, very important as an engineering course. So, it is practice, practice, practice, practice design, practice specification, practice coding. And also remember that to have a good hold over the database, you must be strong in algorithm, data structure and discrete structures, at least the fundamentals of that.

And beyond that, if you need any help, we will be happy to help you, wish you all the best in this course, and in your future career. Thank you very much for being with me in this course. Have a good day and a good time ahead.